Unit 3 Quiz: Two-Way Tables KEY

Name _____

Show your work.

3.1 - Two-Way Tables and Relative Frequency

1. **I read a two-way table, and I fill in missing numbers in it.**

   100 students in 9th and 10th grade chose their favorite sport. Each student chose one sport.

   (a) Fill in the missing numbers. Show your work below the table.

	Soccer	Basketball	Volleyball	Total
9th grade	24	17	11	52
10th grade	21	15	12	48
Total	45	32	23	100

   10th grade, soccer: $45 - 24 = 21$ 10th grade, volleyball: $23 - 11 = 12$

   9th grade total: $100 - 48 = 52$ 9th grade, basketball: $52 - 24 - 11 = 17$

   Basketball total: $17 + 15 = 32$

   Check: $21 + 15 + 12 = 48$ and $45 + 32 + 23 = 100$

   (b) How many students are 10th graders who chose soccer?

   21 students

   (c) How many students did **not** choose volleyball?

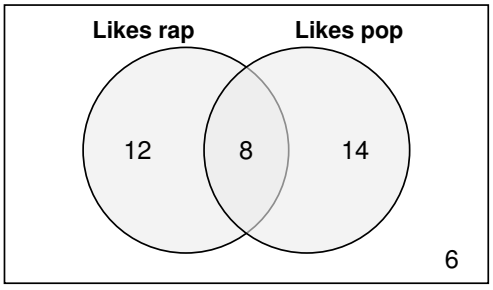
   $100 - 23 = 77$ students

   (d) How many 9th graders did **not** choose soccer?

   $52 - 24 = 28$ students

2. **I move between a Venn diagram and a two-way table, and I make a relative frequency table.**

This Venn diagram shows the music that 40 students like.



(a) Make a two-way table from the Venn diagram.

	Pop	No pop	Total
Rap	8	12	20
No rap	14	6	20
Total	22	18	40

(b) Make a relative frequency table. Divide each count by 40. Write percents.

	Pop	No pop	Total
Rap	$\frac{8}{40} = 20\%$	$\frac{12}{40} = 30\%$	$\frac{20}{40} = 50\%$
No rap	$\frac{14}{40} = 35\%$	$\frac{6}{40} = 15\%$	$\frac{20}{40} = 50\%$
Total	$\frac{22}{40} = 55\%$	$\frac{18}{40} = 45\%$	100%

(c) What percent of all the students like at least one of the two kinds of music?

$12 + 8 + 14 = 34$ students. $\frac{34}{40} = 85\%$ (Or: $100\% - 15\% = 85\%$)

(d) Which statement is supported by the relative frequency table? Circle the letter.

A 8% of the students like both rap and pop.

Ⓑ 55% of the students like pop.

C 30% of the students like neither rap nor pop.

D 35% of the students like rap but not pop.

3.2 - Marginal and Conditional Distributions

3. **I find a marginal distribution and a conditional distribution.**

80 students in 10th and 11th grade chose their favorite school subject.

	Math	**Science**	**Art**	**Total**
10th grade	12	9	9	30
11th grade	18	21	11	50
Total	30	30	20	80

(a) Find the marginal distribution of favorite subject.

Divide each total by the grand total, 80.

Math $\frac{30}{80} = 37.5\%$ Science $\frac{30}{80} = 37.5\%$ Art $\frac{20}{80} = 25\%$

(b) Find the conditional distribution of favorite subject for 11th graders.

Divide by the 11th grade total, 50.

Math $\frac{18}{50} = 36\%$ Science $\frac{21}{50} = 42\%$ Art $\frac{11}{50} = 22\%$

(c) Is each statement a **marginal** or a **conditional** percent? Circle one.

25% of all the students chose art. (Marginal) Conditional

Of the 10th graders, 40% chose math. Marginal (Conditional)

4. **I choose the right total to divide by, and I decide if two categorical variables are associated.**

A student recorded the weather and the school bus for 75 school days.

	Bus late	**Bus on time**	**Total**
Rain	9	6	15
No rain	12	48	60
Total	21	54	75

(a) On what percent of the rainy days was the bus late?

Divide by the rain total, 15. $\frac{9}{15} = 60\%$

(b) On what percent of the days with no rain was the bus late?

Divide by the no rain total, 60. $\frac{12}{60} = 20\%$

(c) Is rain associated with a late bus? Explain.

Yes. The percents are different. The bus was late on 60% of the rainy days, but on only 20% of the days with no rain.

(d) Of the days when the bus was late, what percent had rain? Which fraction gives the answer? Circle the letter.

A $\frac{9}{15}$ (B) $\frac{9}{21}$ C $\frac{9}{75}$ D $\frac{21}{75}$